

A 500 Gbps PAM4 Silicon Microdisk Modulator With a 3.2 THz FSR and >110 GHz Bandwidth

Shenlei Bao^(1,2,†), Houyou Lai^(1,2,†), Chao Cheng^(1,2), Wenfu Zhang^(1,2), Binhao Wang^(1,2)

⁽¹⁾ The State Key Laboratory of Ultrafast Optical Science and Technology, Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an 710119, China, wangbinhao@opt.ac.cn

⁽²⁾ The University of Chinese Academy of Sciences, Beijing 101408, China,

^(†) These authors contributed equally.

Abstract We demonstrate a microdisk modulator with an ultra-high bandwidth exceeding 110 GHz at a 6 dB insertion loss (IL) detuning and a free spectral range (FSR) of 3.2 THz. Clear PAM4 eye diagrams at 448 and 500 Gbps are obtained, with TDECQ values of 1.41 dB and 2.46 dB, respectively.

Introduction

The rapid development of artificial intelligence (AI) has driven an explosive growth in data interconnect demands, exposing the severe bandwidth and power consumption bottlenecks of traditional electrical interconnects. To address these challenges, co-packaged optics (CPO) and optical I/O (OIO) have been proposed as highly effective chip-to-chip interconnect paradigms. Benefiting from excellent compatibility with complementary metal-oxide-semiconductor (CMOS) processes [1], silicon photonics (SiPh) technology is widely recognized as a key enabling technology for CPO and OIO [2-4]. As one of the core components, silicon optical modulators have emerged as a research hotspot, with performance developing rapidly in recent years [5-7].

Among these modulators, microdisk modulators (MDMs) and microring modulators (MRMs) offer high integration densities and inherent wavelength division multiplexing (WDM) capabilities. As the radius shrinks to increase the FSR, sidewall scattering loss and bending radiation loss rise significantly, preventing further

miniaturization without severely degrading the quality (Q) factor. In contrast, MDMs are intrinsically free from inner-sidewall scattering loss. By eliminating the inner etched boundary, MDMs can achieve a much smaller radius than MRMs under identical fabrication processes. This structural advantage enables MDMs to realize a larger FSR, which is essential for accommodating more WDM channels in high-capacity systems.

However, the modulation speeds of current MDMs remain relatively low, which significantly hinders their practical application in future high-speed WDM systems. Conventional MDM designs typically adopt vertical junctions to achieve high modulation efficiency [6]. While these structures provide strong optical-electrical overlap, they also introduce a large junction area, leading to substantial parasitic capacitance. This, in turn, severely limits the electro-optic (EO) bandwidth and overall modulation speed of MDMs.

In this paper, we demonstrate an ultra-high bandwidth (>110 GHz @IL=6 dB) MDM with an FSR of 18.2 nm (3.2 THz), enabled by a novel lateral doping structure that minimizes electrical

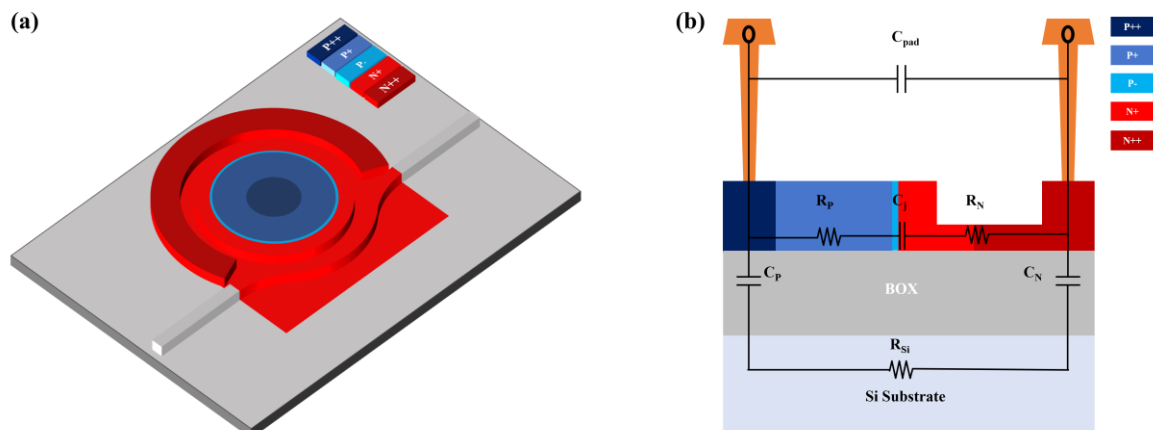


Fig. 1: (a) Schematic of the MDM on an SOI platform. (b) Cross-sectional view of the junction and the corresponding parasitic model of the MDM.

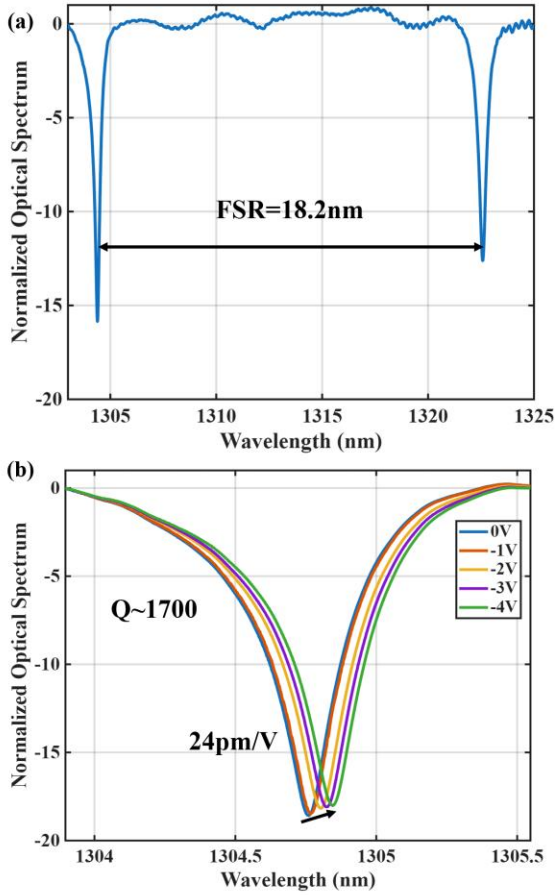


Fig. 2: (a) Measured transmission spectrum of the MDM. (b) Measured tuning efficiency of the MDM.

parasitics and fully compatible with standard silicon photonics fabrication processes. At 448 Gbps PAM4 and 500 Gbps PAM4, transmitter and dispersion eye closure quaternary (TDECQ) values of 1.41 dB and 2.46 dB are achieved, respectively, using a 9-tap FFE at a symbol error rate (SER) of 2×10^{-2} .

Design and Characterization

Fig. 1(a) depicts the schematic diagram of the proposed MDM. The microdisk is designed with a radius of $4.1 \mu\text{m}$ to achieve a large FSR, meeting the channel capacity requirements of 8×400 Gbps or 16×200 Gbps WDM systems. To ensure sufficient coupling for critical coupling of the fundamental mode while effectively suppressing higher-order modes, a bent rib waveguide with a width of 380 nm is employed. Both the microdisk and the bus waveguide share a total silicon thickness of 220 nm and a partial etch depth of 130 nm.

Fig. 1(b) shows the cross-sectional junction diagram of the proposed MDM. The PN junction consists of P⁺, P⁻, and N⁺ regions. The doping concentrations of the P⁺ and N⁺ layers are

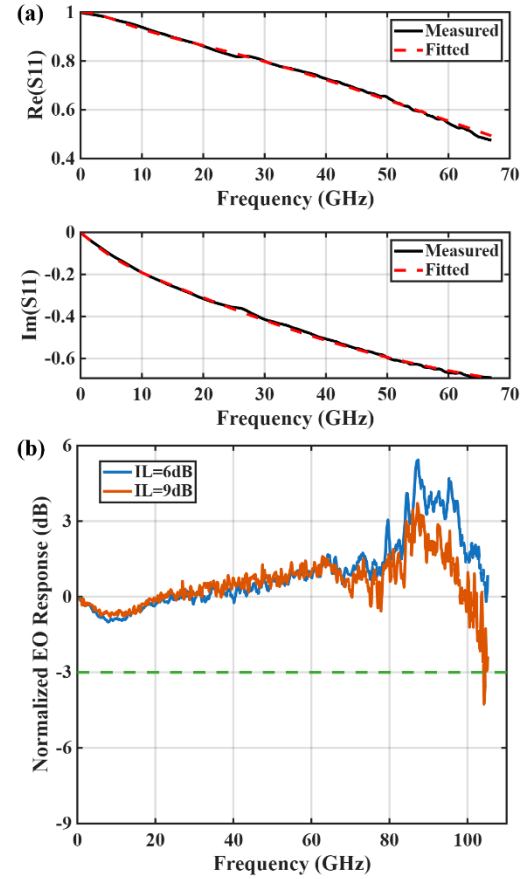


Fig. 3: (a) Measured and fitted S_{11} curves. (b) Measured EO responses at 6 dB and 9 dB insertion loss detuning from maximum transmission power of the MDM under a -3 V bias voltage.

chosen to balance the trade-off between the Q-factor and modulation efficiency. Meanwhile, the P⁻ layer is a thin, lightly doped region intentionally inserted between the heavily doped regions to reduce junction capacitance. At 0 V bias, the depletion region forms at the P⁻/N⁺ interface. As the reverse bias increases, the depletion region extends rapidly into the P⁻ side due to its low doping concentration. Once the P⁻ layer is fully depleted, the structure effectively behaves as a conventional P⁺/N⁺ junction. Therefore, by optimizing the junction position and the width of the P⁻ layer, it is possible to maintain modulation efficiency comparable to that of a conventional P⁺/N⁺ microdisk, while achieving a wider depletion region (lower electrical parasitics) across the operating bias range.

The measured transmission spectrum in Fig. 2(a) exhibits an ultra-wide FSR of 18.2 nm, with higher-order modes effectively suppressed. This large FSR can accommodate 8 WDM channels with 400 GHz spacing or 16 DWDM channels with 200 GHz spacing. The optimized doping profile results in a modulation efficiency of 24 pm/V and a Q-factor of approximately 1700, as shown

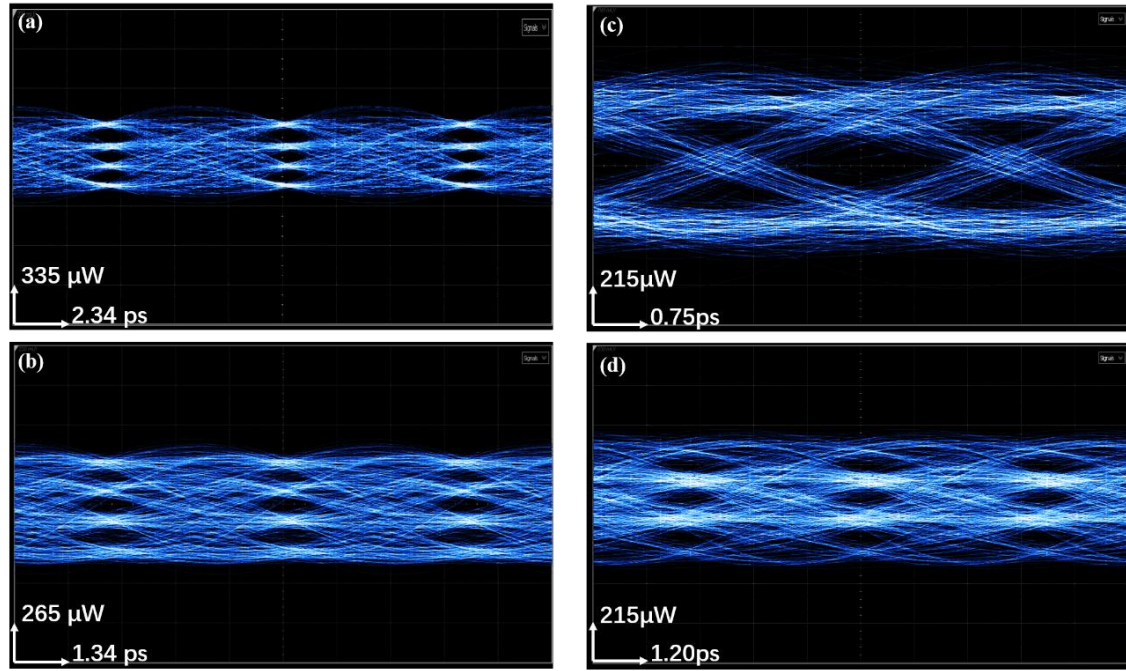


Fig. 4: Measured eye diagrams at data rates of (a) 256 Gbps PAM4, (b) 448 Gbps PAM4, (c) 250 Gbps NRZ and (d) 500 Gbps PAM4.

in Fig. 2(b). The electrical S_{11} parameter of the MDM was measured at a bias voltage of -3 V using a vector network analyser (VNA). Based on the parasitic model [8] and the fitting of the S_{11} curve shown in Fig. 3(a), the extracted RC-limited bandwidth is 298 GHz. Due to the reduced electrical parasitics and the relatively low cavity Q-factor, the measured results in Fig. 3(b) demonstrate an ultra-high EO bandwidth exceeding 110 GHz at 6 dB IL detuning and 104 GHz at 9 dB IL detuning under a -3 V bias voltage.

A 256 GSa/s arbitrary waveform generator (AWG) was used to deliver a 2.5 V_{pp} driving signal to the MDM. The frequency response of the radio frequency (RF) components, including cables and bias-tee, was carefully calibrated by the AWG to compensate for insertion loss and phase distortion. The modulated optical signal, initially around -8 dBm, was amplified to 1 dBm using a praseodymium-doped fiber amplifier (PDFA) before being detected by the oscilloscope, thereby compensating for the coupling loss between the SiPh chip and the fiber. Benefiting from the ultra-high EO bandwidth, Fig. 4(a) and Fig. 4(b) present the 256 Gbps NRZ and 448 Gbps PAM4 eye diagrams, achieving TDECQ values of 0.46 dB and 1.41 dB, respectively, with a 9-tap FFE, at a SER of 2×10^{-2} . These results validate the MDM's potential for 3.2 Tbps WDM systems (16×200 Gbps or 8×400 Gbps). To further explore the MDM's performance limits, a 250 Gbps NRZ eye diagram and a 500 Gbps PAM4 eye diagram are also demonstrated, as shown in Fig. 4(c) and Fig.

4(d). The 500 Gbps PAM4 eye achieves a TDECQ of 2.46 dB with a 9-tap FFE at a SER of 2×10^{-2} .

Conclusions

We have demonstrated a silicon MDM with a 3.2 THz FSR and an ultra-high bandwidth exceeding 110 GHz. To the best of our knowledge, it is the first reported MDM operating at 500 Gbps per wavelength. This result verifies the strong potential of MDMs for next-generation high-speed optical interconnects and optical I/O applications.

References

- [1] S. Shekhar, S. Shekhar, W. Bogaerts, L. Chrostowski, J. E. Bowers, M. Hochberg, R. Soref, B. J. Shastri, "Road-mapping the next generation of silicon photonics," *Nature Communications* 15, 751 (2024), DOI: [10.1038/s41467-024-44750-0](https://doi.org/10.1038/s41467-024-44750-0)
- [2] S. K. Yeh, C. T. Shih, F. Yuan, C. M. Hung, C. H. Chu, H. Y. Lu, J. H. Yang, W. S. Lo, C. H. Chen, S. Y. Tsai, W. C. Tai, S. C. Liu, S. D. Wang, K. Q. Wen, W. C. Wang, C. C. Tung, B. T. Lin, F. Hu, P. C. Yeh, C. H. Huang, T. H. Wu, C. C. Chu, W. C. Kuo, C. Y. Tsai, S. W. Chang, E. C. Chen, C. W. Chiang, Y. M. Wang, F. C. Huang, S. M. Wu, J. Y. Lin, C. T. Tang, W. K. Liu, T. L. Hsieh, W. J. Mao, W. T. Lo, C. Y. Peng, S. H. Su, F. Tsui, N. Shi, V. Shih, and S. F. Huang, "Silicon Photonics Platform for Next Generation Data Communication Technologies," *2024 IEEE International Electron Devices Meeting (IEDM)*, San Francisco, CA, USA, 2024, pp. 1-4, DOI: [10.1109/IEDM50854.2024.10873369](https://doi.org/10.1109/IEDM50854.2024.10873369)
- [3] S. Bao, C. Cheng, X. Bu, H. Lai, X. Yu, J. Zhou, J. Xue, W. Zhang, B. Wang, "Silicon photonic microring-based eight-channel wavelength-division multiplexing transceiver for high-density optical interconnects," *Advanced*

Photonics Nexus 5, 036002 (2026),
DOI: [10.1117/1.APN.5.3.036002](https://doi.org/10.1117/1.APN.5.3.036002)

- [4] J. Xue, Q. Ma, S. Chen, C. Cheng, S. Bao, W. Zhang, B. Wang, "A 64Gbaud/s Hybrid Integrated Silicon Photonic Transceiver with Co-Designed CMOS Driver and TIA for In-Package Optical I/O," *European Conference on Optical Communication (ECOC)*, Frankfurt, Germany, 2024, pp. 372-375.
- [5] C. Han, Q. Yang, J. Qin, Y. Zhou, Z. Zheng, H. Wang, Y. Sun, J. Lu, Y. Wang, Z. Ge, Y. Wu, L. Wang, Z. He, S. Yu, W. Hu, C. Peng, H. Shu, J. E. Bowers, X. Wang, "Exploring 400 Gbps/λ and beyond with AI-accelerated silicon photonic slow-light technology," *Nature Communications* 16, 6547 (2025),
DOI: [10.1038/s41467-025-61933-5](https://doi.org/10.1038/s41467-025-61933-5)
- [6] A. Novick, S. Wang, A. Rizzo, V. Gopal, K. Bergman, "Ultra-Efficient Interleaved Vertical-Junction Microdisk Modulator with Integrated Heater," *2024 Optical Fiber Communications Conference and Exhibition (OFC)*, San Diego, CA, USA, 2024, pp. 1-3,
DOI: [10.1364/OFC.2024.W2A.21](https://doi.org/10.1364/OFC.2024.W2A.21)
- [7] J. Xue, S. Bao, C. Cheng, W. Zhang, B. Wang, "An 8×256 Gbps Silicon Photonic DWDM Transmitter with Thermally Stable Microring Modulators," *2025 Optical Fiber Communications Conference and Exhibition (OFC)*, San Francisco, CA, USA, 2025, pp. 1-3,
DOI: [10.1364/OFC.2025.Th1E.3](https://doi.org/10.1364/OFC.2025.Th1E.3)
- [8] S. Bao, Y. Ma, J. Xue, J. Wu, N. Qi, B. Wang, "A Comprehensive Equivalent Circuit Model of Silicon Microring Modulators for Photonics-Electronics Codesign," *Journal of Lightwave Technology*, 42(14), 4924-4932 (2024),
DOI: [10.1109/JLT.2024.3382091](https://doi.org/10.1109/JLT.2024.3382091)